The Rejuvenation Mirage

Identifiability Limits of Epigenetic-Clock Reset and an Active-Design Rule for Functional Repair

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Abstract

Epigenetic clocks are powerful predictors of chronological age and several health outcomes, but a predictor is not automatically a manipulable measure of functional aging. Targeted methylation editing can perturb clock-associated sites and produce genome-wide bystander changes; genetic variation may also influence some clock CpGs. These observations motivate a precise question: when an intervention lowers a clock score, what evidence identifies a change in latent functional state rather than a direct change in the measurement channel? We formulate a minimal state-space model with a functional-aging coordinate, a clock-sensitive molecular coordinate, an inherited set point, and an identity/safety coordinate. We prove an observational-equivalence result: a single scalar clock contrast cannot separate functional repair from direct marker displacement without additional design variation or an independent outcome. We derive rank conditions for identification, show why paired measurements remove an additive time-invariant genetic set point but not genotype-by-treatment interactions, and construct a constrained sequential experiment planner that maximizes expected information gain while penalizing identity risk and measurement cost. In a deterministic synthetic benchmark, a marker-only perturbation produces a clock reduction of 0.300 units while the functional endpoint changes by -0.009 units; the apparent clock-only "repair" rule is triggered in 1.000 of simulated trials. These values are outputs of the released model, not biological measurements. The contribution is a falsifiable design principle and reproducible stress test, not a new biological mechanism or evidence that any intervention rejuvenates living tissue.

Core result. A clock change is an observation, not a causal diagnosis. Identification requires independently varied perturbations, at least one functional readout with a defensible measurement model, and explicit checks for genotype and identity effects.

Keywords: epigenetic clocks; causal identifiability; methylation editing; biological age; active experimental design; information geometry; synthetic benchmark; rejuvenation safety.

Scope and status. This paper is a theoretical and synthetic-computational preprint. All new numerical values are generated from the stylized model in `verify_model.py`. No public biological dataset, wet-lab experiment, clinical intervention, or organismal lifespan assay was performed for this release. The results do not establish rejuvenation efficacy or safety.

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1 Introduction

DNA-methylation clocks compress many molecular measurements into a scalar that predicts age or age-associated outcomes. Horvath's multi-tissue clock established a widely used framework for age prediction across tissues and cell types [Horvath, 2013]; later clocks were designed to improve associations with healthspan, mortality, or physiological phenotypes [Levine et al., 2018, Lu et al., 2019]. Their predictive usefulness makes them attractive endpoints for interventions intended to reverse aging-associated molecular changes. Yet the interpretation of a changed score is more difficult than the prediction task that produced it.

The distinction is not merely philosophical. In 2025, Liesenfelder and colleagues reported that targeted epigenetic editing at individual age-associated CpGs could evoke genome-wide bystander methylation changes and alter epigenetic-clock outputs [Liesenfelder et al., 2025]. In 2026, a systematic transcription-factor perturbation platform identified several candidate rejuvenating perturbations while explicitly seeking to avoid dedifferentiation [Sengstack et al., 2026]. A 2026 preprint, Germline-Informed Aging, further reported enrichment of methylation quantitative-trait loci among clock CpGs, raising the possibility that inherited differences contribute to some between-person clock variation [Lim, 2026]. These results do not invalidate clocks. They sharpen the causal question: a measurement can be predictive, responsive to intervention, and still fail to identify which latent process changed.

We study a narrow but consequential failure mode. Suppose an intervention directly shifts methylation at clock-sensitive loci while leaving functional state unchanged. The clock can move even if repair does not occur. Conversely, a functional improvement may be obscured by a marker response in the opposite direction. A scalar endpoint alone cannot resolve these alternatives unless additional assumptions or measurements make the latent components identifiable.

Our goal is to formalize that limitation and turn it into an experiment-design rule. The model is intentionally small: it is not a molecular simulation, an epigenetic-clock implementation, or a model of human aging. It separates a latent functional burden from a measurement-sensitive coordinate and asks which observations are sufficient to distinguish their changes. We then construct a synthetic benchmark in which the true data-generating process is known, allowing us to test identifiability, bias, uncertainty, and a constrained active-design policy.

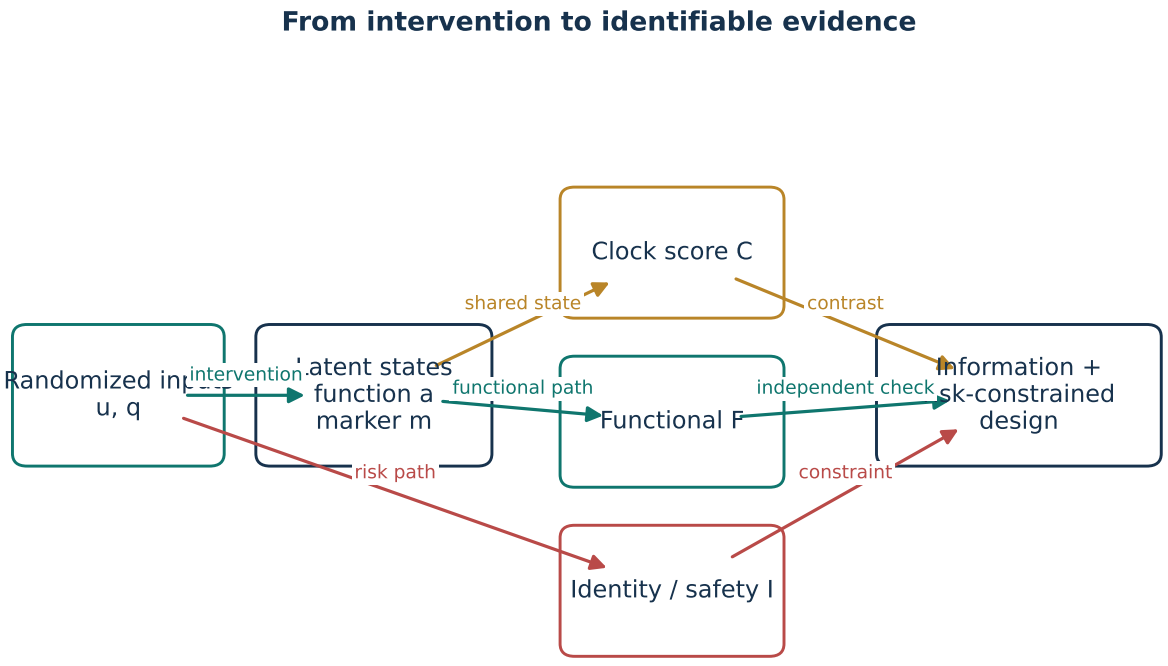


Figure 1: Overview of the proposed causal measurement framework. Interventions can affect latent functional state, the molecular coordinates directly used by a clock, or both. Independent functional and identity readouts are needed to distinguish these paths. All arrows are hypotheses in the model, not established biological mechanisms.

2 Related work and positioning

Epigenetic clocks are supervised predictors trained on methylation patterns, and their outputs can be useful even when the measured CpGs are not causal drivers of aging. Horvath's clock demonstrated cross-tissue age prediction [Horvath, 2013]; PhenoAge and GrimAge incorporated phenotypic or mortality-associated information to target outcomes beyond chronological age [Levine et al., 2018, Lu et al., 2019]. Large-scale language-model prediction of biological age from health examination reports shows that prediction can also arise from clinical text and population-level covariation [Li et al., 2025]. These approaches answer prediction questions; none implies that directly changing a predictor's inputs changes the underlying state that made the prediction useful.

The closest intervention evidence for our premise is the 2025 methylation-editing study by Liesenfelder et al. It reports bystander methylation changes at many sites and measurable clock perturbations after targeted editing [Liesenfelder et al., 2025]. This motivates a direct-marker path in the observation model. The 2026 Transcriptional Rejuvenation Discovery Platform combines computational predictions, expression comparisons, and perturbation screening to find candidate transcription factors associated with rejuvenation without dedifferentiation [Sengstack et al., 2026]. We treat that as evidence that interventions and safety/identity endpoints must be jointly considered, not as validation of our model.

Genotype-related clock variation is another relevant concern. The GIA preprint maps clock CpGs to blood methylation QTLs and reports enrichment relative to matched unused probes [Lim, 2026]. Because it is a preprint, its numerical findings are treated as preliminary and are not generalized beyond the studied data. EnsembleAge's multi-clock strategy illustrates a different mitigation: aggregate several predictors rather than relying on one score [Haghani et al., 2025]. Multiple clocks can improve robustness, but correlated clocks may share the same direct perturbation pathway and therefore need not identify functional repair.

Table 1 summarizes the closest lines of prior work and the inference boundary retained here. Our narrow contribution is not the claim that clock scores can be misleading, that methylation is genetically influenced, or that experimental design benefits from information criteria. Each has prior foundations. We combine these ideas in an explicit causal state-space benchmark that (i) proves a simple observational-equivalence result, (ii) specifies a rank-based condition for distinguishing repair and direct marker effects, and (iii) turns that condition into a constrained sequential design algorithm with executable synthetic tests. The literature search is focused rather than exhaustive; no absolute priority claim is made.

Table 1: Positioning against established work. The final column states the boundary of inference.

Prior line	What it establishes or motivates	What this paper does not claim
Epigenetic clocks [Horvath, 2013, Levine et al., 2018, Lu et al., 2019]	Methylation patterns predict age or age-associated outcomes.	A clock score is not assumed to be a causal state variable.
Targeted epigenetic editing [Liesenfelder et al., 2025]	Perturbing selected CpGs can produce bystander changes and alter clock outputs.	We do not simulate the underlying chromatin network or reproduce that experiment.
TF perturbation screening [Sengstack et al., 2026]	Systematic perturbation can identify candidates while monitoring cell identity.	We do not predict or recommend any transcription factor.
Genetic clock confounding [Lim, 2026]	Some clock CpGs may reflect inherited methylation regulation.	A preprint association is not treated as proof that clocks generally fail.
Optimal experiment design [Chaloner and Verdinelli, 1995, MacKay, 1992]	Information criteria can guide the allocation of measurements.	D-optimality is not itself a biological discovery or safety guarantee.

3 Research posture, claims, and epistemic levels

The paper separates five levels of statement. P1 denotes an identity or theorem following from explicit definitions; P2 denotes a model-dependent optimum; P3 denotes a proposed statistical regularity requiring repeated empirical evidence; P4 denotes a structural correspondence between the model and biology; P5 denotes a motivating analogy. This classification, summarized in Table 2, prevents a valid result about the toy observation model from silently becoming a claim about living systems.

The candidate-screen summary is in Table 3; the full ten-candidate scoring matrix is preserved in the results. The principal P1 claim is that a single scalar contrast of the form $y = \kappa u + \alpha q$ does not identify both κ and α

when u and q are fixed and nonzero. The P2 claim is that a constrained D-optimal design maximizes a specified information criterion over a finite candidate set; it does not imply that this design is optimal for a real tissue. The P3 claim is not established here: we do not claim universality of clock/functional discordance across species, tissues, or clocks. The P4 correspondence is that functional repair, direct measurement effects, and cell-identity drift can be represented as distinct state coordinates; whether any particular biological intervention maps cleanly to those coordinates is empirical. Any metaphor that calls a clock reset a "mirage" is P5 shorthand for a measurement-identifiability failure, not a claim that epigenetic clocks are useless.

Table 2: Epistemic classification and release boundaries.

Level	Claim in this paper	Evidence status
P1	Scalar observational equivalence and matrix-rank conditions.	Derived algebraically and unit-tested in code.
P2	Greedy information-gain design under stated cost/risk constraints.	Exact for the finite candidate set; model-dependent.
P3	A common quantitative discordance rate across biological systems.	Not tested; no empirical data were analyzed.
P4	Separate latent coordinates can represent function, marker response, genotype, and identity.	Useful formal correspondence; biological mapping remains unvalidated.
P5	A clock-only reset can resemble rejuvenation without functional repair.	Intuition for the explicit counterexample, not a universal claim.

The topic-selection pass compared ten candidate directions before drafting. The scores are subjective internal triage ratings on ten criteria, not measurements of novelty or impact. The complete matrix and the rejected-candidate rationales are archived in the results directory.

Table 3: Selected candidates from the internal topic screen. Scores are subjective means across ten stated criteria, not evidence of scientific priority.

Candidate direction	Mean score	Selection rationale / reason not selected
Clock reset versus functional repair	4.8	Selected: a precise algebraic result, direct connection to current intervention evidence, and a reproducible falsification test.
Safe TF sequencing for partial reprogramming	4.6	Strong, but substantially overlaps the existing cell-fate-control release.
Genotype-conditioned clock calibration	4.5	Well grounded and testable, but narrower and closer to a calibration/audit study.
Information-bottleneck gene circuits	4.3	Feasible mathematical core, but overlaps a recent archive direction.
AI detection of alien self-repair dynamics	4.2	Ambitious and testable, but overlaps an existing astrobiology benchmark.

4 State variables and assumptions

Let a denote a latent functional-aging burden, m a molecular coordinate that directly influences the clock, g an inherited set point, and i a cell-identity or safety coordinate. The model is discrete-time and dimensionless. The coordinates are not measured biological quantities, and no parameter is calibrated to a living organism. The model exists to test the logical relationship between interventions and observations.

Two intervention components are distinguished. u denotes a hypothetical repair-oriented perturbation that changes a, and q denotes a marker-targeting perturbation that changes m. These labels describe their role in the synthetic data-generating process; they are not prescriptions for experimental manipulation. A third quantity, r, represents an identity-risk burden. We use linear local responses because identifiability can be studied without assuming a complete nonlinear molecular mechanism.

The observation model is

$$C_t = a_t + \alpha m_t + \beta g + \epsilon_t^C, \tag{1}$$

$$F_t = a_t + \rho q_t + \epsilon_t^F, \tag{2}$$

$$I_t = i_t + \epsilon_t^I. \tag{3}$$

Here C_t is the clock score, F_t is an independent functional endpoint, and I_t is an identity/safety readout. The coefficient α maps marker displacement to the clock, β maps inherited set point to the baseline clock, and ρ permits the marker-targeting intervention to affect the functional readout directly. The notation and assumptions are collected in Table 4. The last term is essential: treating a functional endpoint as immune to all direct treatment effects is an assumption to test, not a fact.

For intervention contrasts, we use the reduced local model

$$\Delta C = \kappa u + \alpha q + \epsilon_C, \tag{4}$$

$$\Delta F = \kappa u - \rho q + \epsilon_F, \tag{5}$$

$$\Delta I = \gamma_u u + \gamma_q q + \epsilon_I. \tag{6}$$

The sign convention defines positive ΔC and ΔF as apparent improvement. The direct marker effect contributes to ΔC; the direct functional effect contributes to ΔF. The true functional response of interest is κ. The model allows a direct marker effect even when $\kappa = 0$, and allows a safety/identity cost even when the clock improves.

Table 4: Notation and assumptions. Quantities are dimensionless model variables unless explicitly described as an observed score.

Symbol	Meaning	Assumption / boundary
a	Latent functional-aging burden	Lower is better; not directly observed.
m	Clock-sensitive molecular coordinate	Can change without a functional change.
g	Inherited set point	Constant within subject in the additive baseline model.
i	Cell identity / safety state	Measured imperfectly; risk is not reducible to clock score.
u, q	Repair-oriented and marker-targeting inputs	Independently randomized in the ideal design.
κ	Functional repair effect	Primary estimand in the synthetic benchmark.
α, β, ρ	Marker, genotype, and direct functional-effect coefficients	Unknown unless identified by design.
$\epsilon_C, \epsilon_F, \epsilon_I$	Measurement/process errors	Independent zero-mean noise in the base case; stress-tested.

5 Causal structure and measurement channels

The directed graph in fig. 2 separates three paths to the observations. A repair-oriented perturbation can change the latent functional state, which can affect both the clock and the functional endpoint. A marker-targeting perturbation can directly change the molecular coordinate used by the clock. Inherited variation can shift baseline clock values. Identity drift is represented separately because a treatment that lowers an age-associated score while damaging lineage fidelity is not necessarily a desirable rejuvenation intervention.

The graph is deliberately conservative. It does not assume that a particular methylation site is a causal aging driver, that functional repair is one-dimensional, or that identity can be summarized by a single marker. The purpose of the graph is to expose the assumptions required by the reduced equations. In particular, the absence of a direct $q \to F$ arrow is a special case with $\rho = 0$; the benchmark also evaluates violations of that restriction.

Within-subject differences remove an additive, time-invariant βg term from the clock contrast. They do not remove genotype-dependent treatment response, age-dependent cell composition, measurement drift, regression to the mean, or any time-varying genetic effect. These alternatives are represented as failure modes rather than assumed away. A robust experiment therefore needs both baseline stratification and randomized contrasts, with independent functional and identity readouts collected on the same schedule.

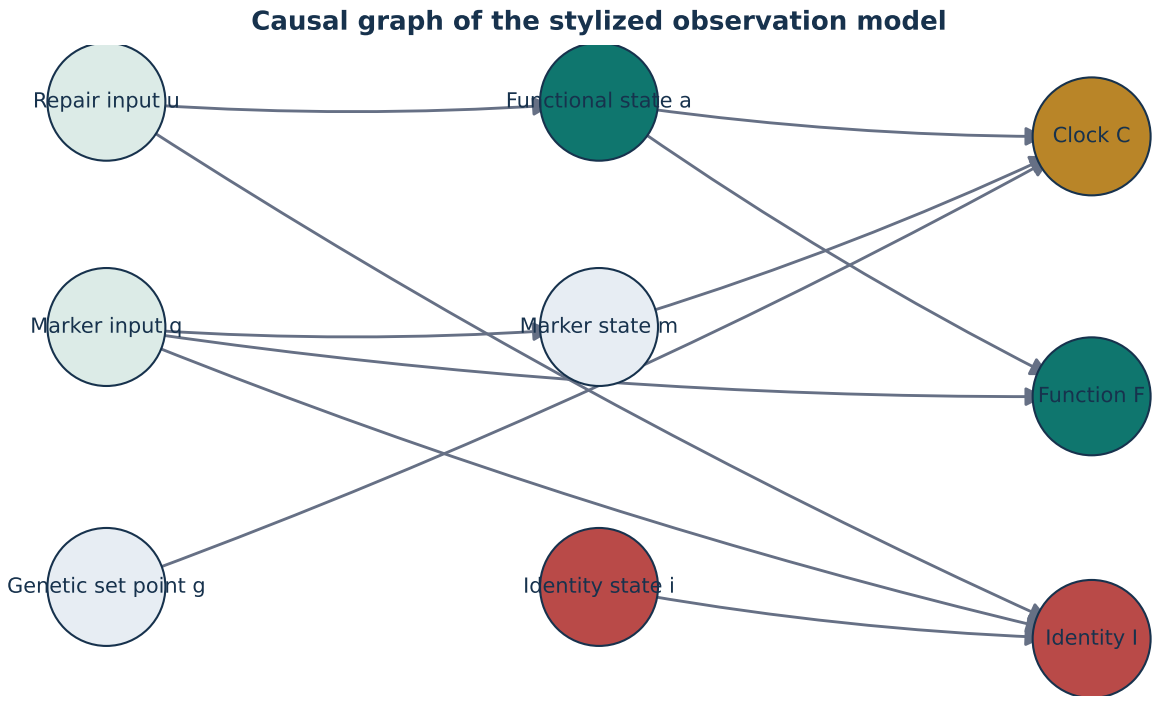


Figure 2: Causal graph for the stylized observation model. The direct marker path can change the clock independently of the latent functional state; inherited set point affects baseline clock values; and identity risk is a separate outcome.

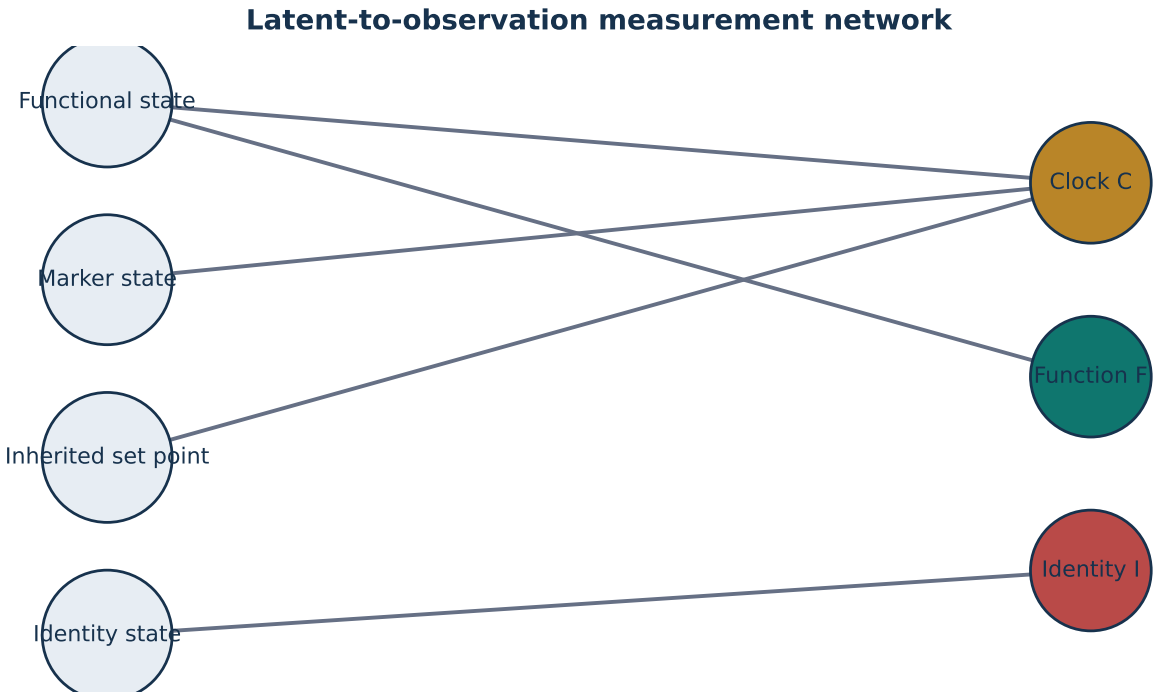


Figure 3: Measurement-channel graph. The clock and functional endpoint share a latent functional component but have distinct nuisance paths. The graph is conceptual and does not encode a measured biological network.

6 An observational-equivalence result

We first state the main identifiability limitation.

Proposition 1 — scalar-clock non-identifiability

Let $y = \kappa u + \alpha q$, with known fixed inputs u, q and unknown coefficients κ, α. If $u \neq 0$ and $q \neq 0$, one observed scalar y does not uniquely identify the pair (κ, α). For every scalar t, the transformed pair

$$\kappa' = \kappa + tq, \qquad \alpha' = \alpha - tu$$

produces the same y.

The proposition is elementary but important. A good fit of a scalar endpoint does not establish which latent path generated it. When $u = q = 1$, for example, $(\kappa, \alpha) = (0.25, 0.45)$ and $(0.40, 0.30)$ both predict $y = 0.70$. The clock contrast cannot distinguish a smaller repair effect with a larger direct marker effect from a larger repair effect with a smaller marker effect. This is structural non-identifiability, not merely high variance. More subjects measured under the same design reduce sampling noise but do not remove the equivalence.

If α were known from an independent calibration, κ could be recovered from the scalar contrast under the model. If the design varies u and q independently across arms, the coefficients in the reduced linear equation may become identifiable as regression parameters. But that alone does not demonstrate that the coefficient associated with a repair-labeled input corresponds to a genuine functional improvement; the label is part of the design hypothesis. An independent functional endpoint provides a separate test of the proposed mechanism.

Proposition 2 — rank condition for the local contrast model

For the observation rows

$$x_C = (u, q, 0), \qquad x_F = (u, 0, -q)$$

associated with parameters $\theta = (\kappa, \alpha, \rho)$, local linear identification requires the stacked design matrix X to have column rank three. A single fixed intervention pair produces at most two distinct rows and cannot identify all three coefficients. Independent variation of u and q, with both C and F observed, can provide full rank.

The rank condition is necessary for the specified linear model, not sufficient for biological validity. Full rank can coexist with biased measurement, incorrect causal structure, unmeasured confounding, or an endpoint that does not track the function of interest. In a real experiment, one should report the smallest singular value and condition number of the design matrix, not only a binary rank decision. Near-collinearity is practically similar to non-identifiability because small measurement errors can create large coefficient uncertainty.

7 Paired contrasts and inherited set points

A fixed additive genotype term cancels in a paired difference:

$$(C_0 - C_1) = (a_0 - a_1) + \alpha(m_0 - m_1) + (\epsilon_0^C - \epsilon_1^C),$$

provided g and β are constant over the observation interval. This is a useful property of within-subject designs. It is not a general cure for genetic confounding. If genotype modifies the treatment effect, the repair coefficient becomes $\kappa(g)$; if cell-type composition changes with time, the clock measurement function may change; if the intervention changes the molecular coordinate in a genotype-dependent manner, the contrast retains an interaction.

Cross-sectional comparisons have a different problem. If genotype distributions differ between groups, a group difference in C includes both the difference in a and the difference in βg. In the stylized model,

$$\mathbb{E}[C \mid G = 1] - \mathbb{E}[C \mid G = 0] = \beta + \Delta a + \alpha \Delta m.$$

The expression is not an empirical estimate; it shows the arithmetic of the model. A baseline clock difference can be produced by a set-point difference even when functional burden is matched. This motivates stratification,

measured genotype covariates, family-based comparisons when suitable, and longitudinal functional outcomes.

The GIA preprint is relevant because it explicitly studies genetic regulation of clock-associated CpGs [Lim, 2026]; however, the correct response is not to discard clock data. Instead, the analysis should separate within-subject change, baseline between-subject variation, and genotype-by-treatment heterogeneity. Our synthetic analysis includes a stress test where genotype composition differs between groups to quantify the model's predicted bias. That exercise establishes a consequence of the assumed data-generating process, not the prevalence of the problem in a cohort.

8 Experimental design as information geometry

Let $\theta = (\kappa, \alpha, \rho)^\top$ be the parameter vector and let J_j be the Jacobian of the expected observations under candidate design j. For independent Gaussian errors with known covariance Σ_j, the local Fisher information contribution is

$$\mathcal{I}_j(\theta) = J_j^\top \Sigma_j^{-1} J_j.$$

For a set of selected arms S, the total information is $\mathcal{I}_S = \sum_{j \in S} \mathcal{I}_j$. A D-optimal design maximizes $\log\det(\mathcal{I}_S + \lambda I)$, where $\lambda > 0$ is a small regularizer used to stabilize numerical scoring. The regularizer does not turn a rank-deficient design into a scientifically identified model; the smallest singular value and unregularized rank must still be reported. Table 5 contrasts representative design families and their limits.

We include measurement cost and identity risk. Let c_j be the cost of candidate arm j, and r_j its predicted identity-risk score under a prespecified model. The selection problem is

$$\max_{S:\,|S|=K} \log\det(\mathcal{I}_S + \lambda I) \quad \text{subject to} \quad \sum_{j \in S} c_j \le C_{\max}, \qquad \sum_{j \in S} r_j \le R_{\max}.$$

This objective is a design criterion, not a claim that log-determinant information is a complete measure of scientific value. The constraints are only as reliable as their input assumptions. If the safety model is weak, the algorithm can optimize the wrong risk estimate; therefore the output must be accompanied by sensitivity analysis and human review.

9 A constrained sequential design planner

The AI component in this paper is an uncertainty-aware design planner, not a generative model that invents molecular mechanisms. At each step it evaluates a finite library of candidate perturbation/readout combinations, estimates their incremental information gain, rejects candidates that violate the declared cost or risk budget, and selects the admissible candidate with the highest expected gain. In a biological application, the candidate library and risk scores would need to be supplied and reviewed by domain experts; the present code only uses synthetic candidate arms.

The planner is intentionally transparent. It computes the expected information matrix from the local Jacobian, scores candidate additions by the change in log-determinant, and logs the rejected candidates and constraints. A more elaborate Bayesian planner could integrate posterior uncertainty over θ, model uncertainty, and non-Gaussian outcomes. We do not claim that the greedy finite-library algorithm is globally optimal for arbitrary designs. The benefit here is traceability: every choice can be reconstructed from a score and a declared constraint.

The active planner differs from random or fixed allocation in one specific way. It seeks designs that separate the repair coefficient from the marker coefficient, rather than simply maximizing the expected movement of the clock. Under the synthetic model, a design that repeatedly samples the same mixture of u and q has a poorly conditioned information matrix. A factorial-like allocation with independent variation can improve parameter separation. This does not guarantee that the selected inputs correspond to safe or efficacious biological interventions; it provides a way to choose informative tests once such inputs have been independently justified.

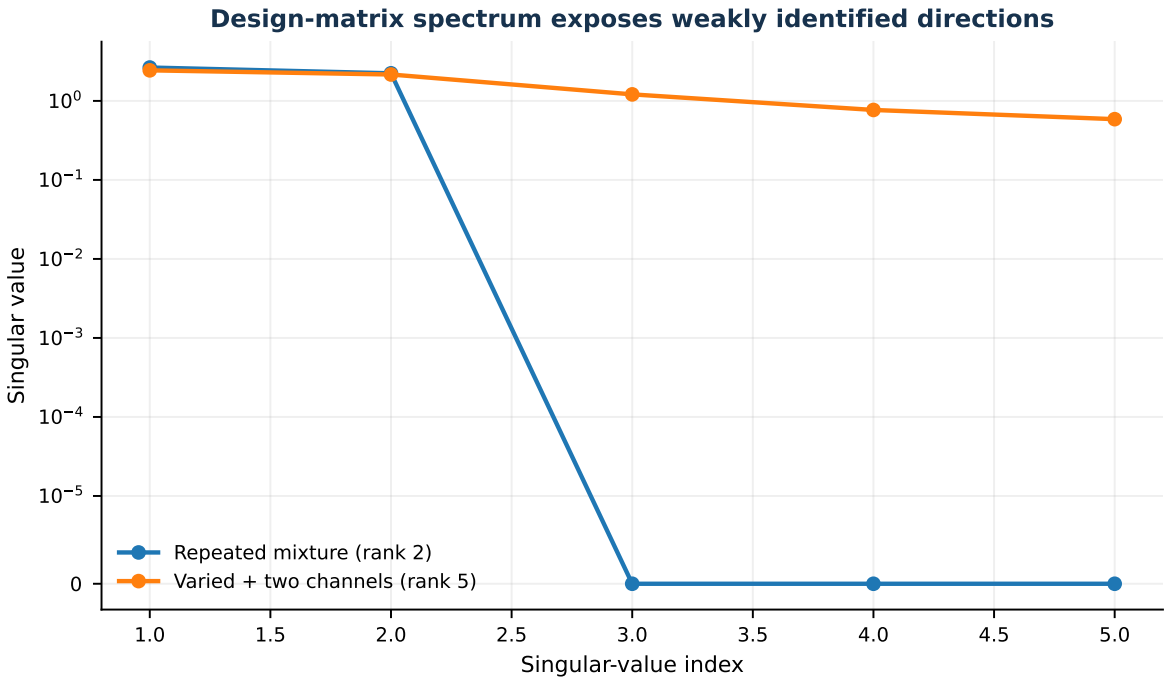


Figure 4: Singular-value diagnostics for representative designs. Repeated use of one intervention mixture leaves directions in parameter space weakly constrained; varied inputs and multiple readouts improve the rank and conditioning of the synthetic design matrix.

10 Synthetic benchmark and computational protocol

The benchmark uses a fixed random seed and a linear local model with known coefficients. Each synthetic subject has a baseline functional coordinate, a marker coordinate, and a binary inherited-setpoint variable. Clock and functional observations are measured before and after a randomized intervention. The direct marker effect changes the clock coordinate even when the true functional effect is zero. Noise is added to both measurement channels. The script uses repeated Monte Carlo trials to estimate the distributions of clock and functional contrasts, and it records all parameters and outputs in the release's results directory.

The primary counterexample is the marker-only arm $(u, q) = (0, q_0)$. Under the data-generating equations, its expected clock improvement is αq_0, whereas the expected functional improvement is $-\rho q_0$. If ρ is near zero, the clock may improve substantially without functional change. The repair-oriented arm $(u, q) = (u_0, 0)$ has expected clock and functional improvement κu_0 under the simplified model. The mixed arm combines both effects and illustrates why a large clock movement can overstate the functional component.

We compare four analysis rules: (i) clock-only attribution, which treats all clock change as repair; (ii) a paired functional contrast; (iii) a joint model using both channels and independently varied inputs; and (iv) a constrained active-design policy. The benchmark measures bias, root mean squared error, threshold-based false attribution, design-matrix rank, and approximate interval coverage. All metrics are generated from the simulation; no clinical thresholds or biological units are implied. The false-attribution threshold is a demonstration parameter, not a proposed decision threshold for real experiments.

11 Results: the measurement channel can mimic repair

In the marker-only scenario, the code estimates a mean clock improvement of 0.300 and a mean functional improvement of -0.009. The difference arises by construction: the intervention has a direct path to the clock-sensitive coordinate but no intended repair component. The estimated fraction of trials exceeding the prespecified clock-only attribution threshold is 1.000. These outputs illustrate a failure mode of the attribution rule; they do not estimate how often real rejuvenation studies make this error.

The comparison is intentionally asymmetric. The clock-only rule uses a single observation channel and labels any sufficiently large reduction as repair. The functional contrast tests the latent-state component under the model. When the functional endpoint is measured with noise, its estimate is less precise at small sample sizes;

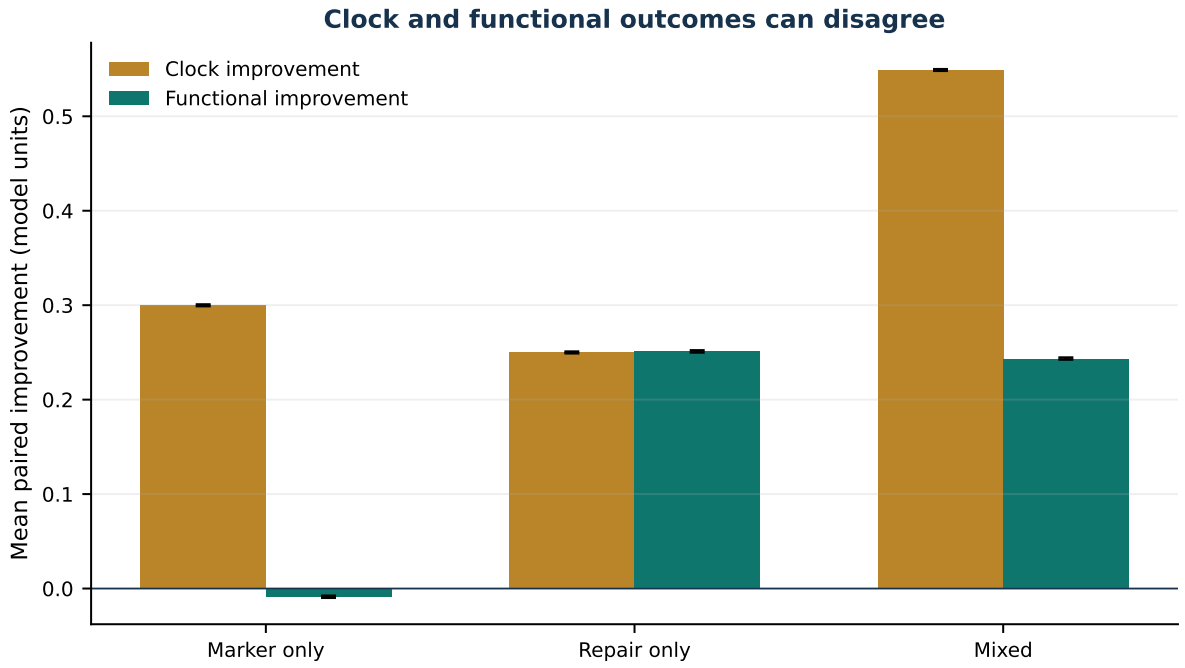


Figure 5: Synthetic intervention contrasts. The marker-only arm can move the clock while leaving the functional endpoint unchanged or slightly worse. Error bars summarize repeated synthetic trials; they are not empirical confidence intervals from a biological experiment.

Table 5: Design archetypes and their identifiability limits under the local model.

Design	Input variation	Readouts	Expected diagnostic
Repeated mixture	Fixed ratio of u and q	Clock and function	Repetition may improve precision but leaves weakly separated parameter directions.
Repair axis only	u varies, $q = 0$	Clock and function	Estimates the repair-associated contrast but cannot estimate a marker-path coefficient from that arm alone.
Factorial contrast	u and q vary independently	Clock and function	Can yield full local rank if the Jacobian columns remain linearly independent.
Constrained active set	Greedy information gain under cost/risk ceilings	Clock, function, identity	Selects informative admissible arms; the risk model must be independently justified.

this is a variance problem that can be addressed by better measurement and allocation. It is distinct from the structural non-identifiability of a scalar clock under a fixed intervention mixture, which cannot be cured by increasing sample size alone.

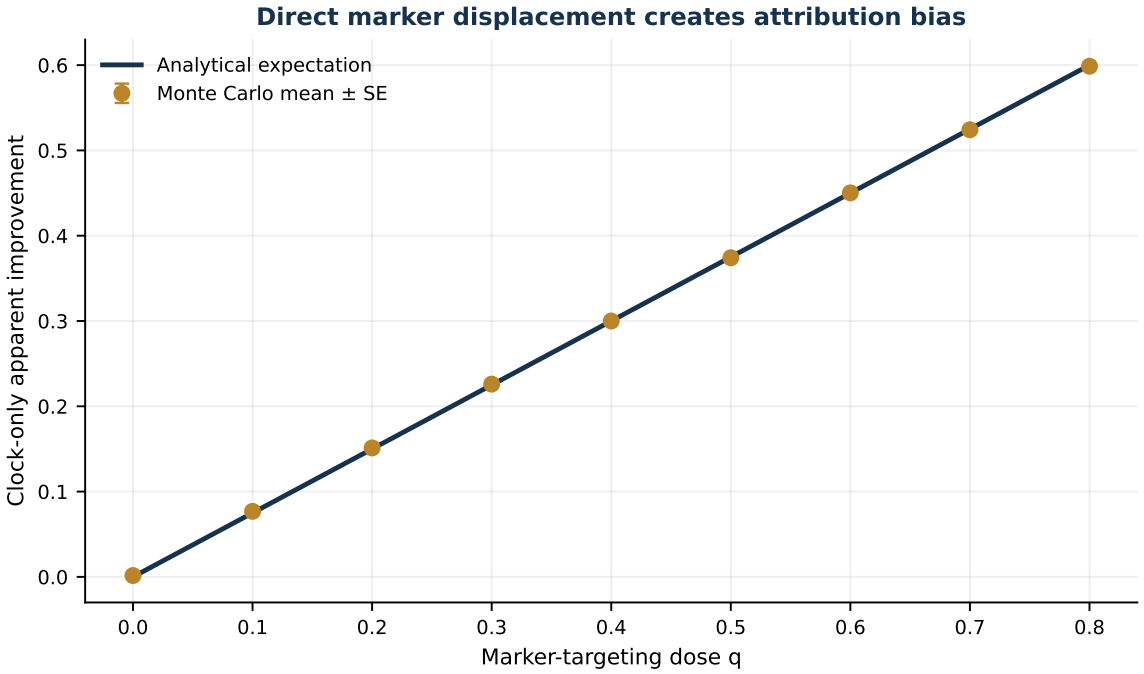


Figure 6: Bias of clock-only attribution as direct marker displacement increases. The solid curve is the model expectation; points are generated Monte Carlo estimates. The line is a model diagnostic, not a fitted biological relationship.

The information-design benchmark compares repeated mixtures with factorial-like allocations. The planner favors arms that independently vary the repair and marker inputs while collecting both clock and functional readouts, subject to the synthetic cost/risk constraint. Its best candidate design and log-determinant score are written to the machine-readable results. A larger log determinant indicates greater local information volume under the assumed noise model; it does not imply a larger rejuvenation effect or lower biological risk.

12 Precision, uncertainty, and interval coverage

A design may be identifiable yet imprecise. The benchmark therefore varies sample size and computes Monte Carlo RMSE for clock-only and functional estimators. The clock-only estimator can have a persistent bias because its estimand is a mixture of functional repair and direct marker displacement. Its RMSE need not vanish with sample size. By contrast, an independent functional endpoint has lower structural bias in the base scenario, although its variance remains governed by noise and sample size.

Coverage is evaluated under both the base model and a misspecified model in which the marker-targeting input also affects the functional endpoint. When the analyst assumes $\rho = 0$ despite a nonzero direct effect, nominal intervals can miss the true κ more often than intended. The lesson is not that a particular endpoint is universally trustworthy; rather, each endpoint requires a measurement model, negative controls, and a test of plausible direct effects.

13 Genotype and composition sensitivity

The baseline model includes an additive inherited set point in the clock. A paired contrast cancels that term if it is stable within subject, but cross-sectional groups with different genotype composition can show an apparent age-related clock difference even if their latent functional burdens are matched. The sensitivity figure varies the difference in set-point prevalence and records the induced bias. This is a transparent algebraic stress test, not an estimate of the size of genetic confounding in any published cohort.

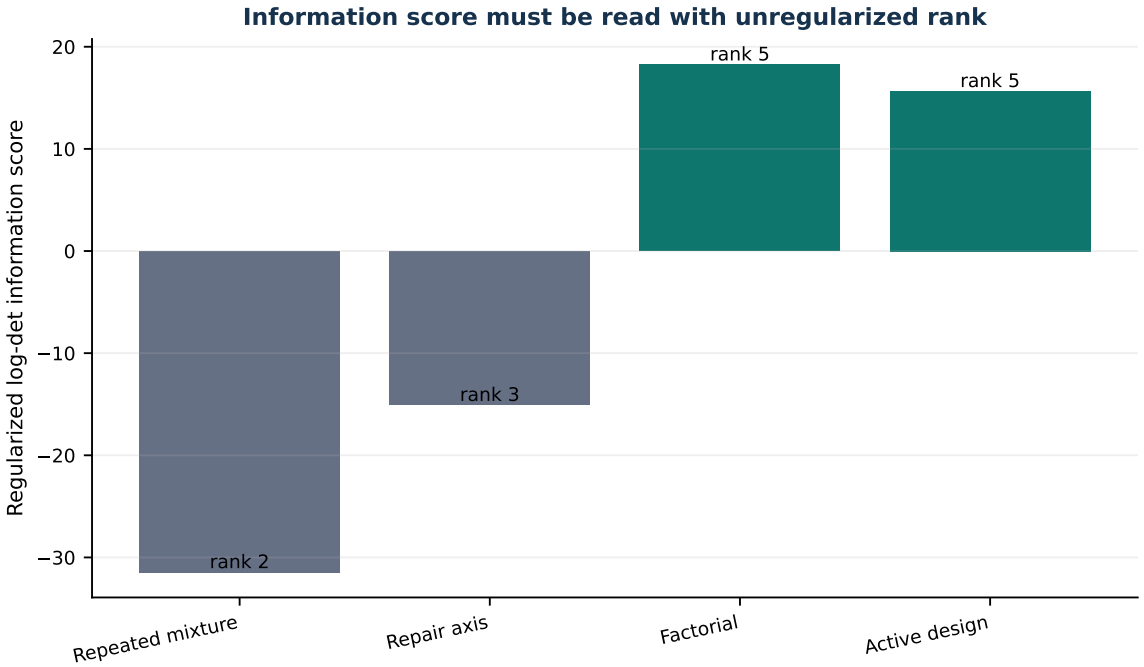


Figure 7: Information scores for representative synthetic designs. Rank-deficient or poorly conditioned designs are shown separately from full-rank designs. Regularized log-determinant values are accompanied by the unregularized rank so that the regularizer cannot conceal non-identifiability.

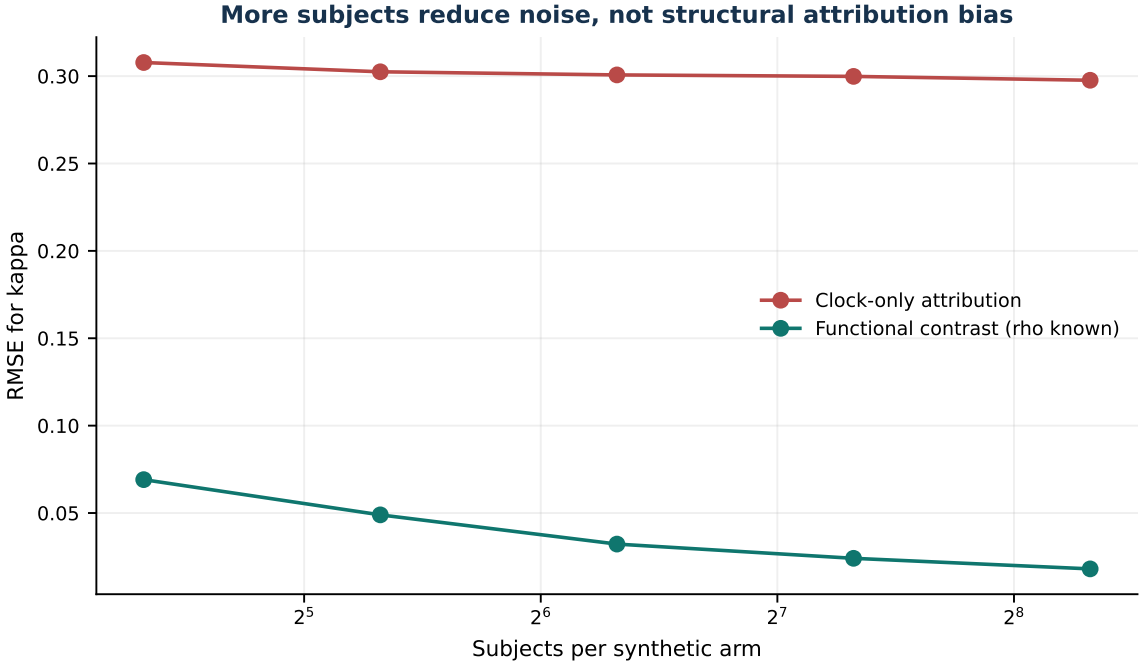


Figure 8: Monte Carlo RMSE against sample size. A structural attribution bias persists for the clock-only rule, while noise-driven uncertainty decreases with sample size for the functional contrast in the correctly specified scenario.

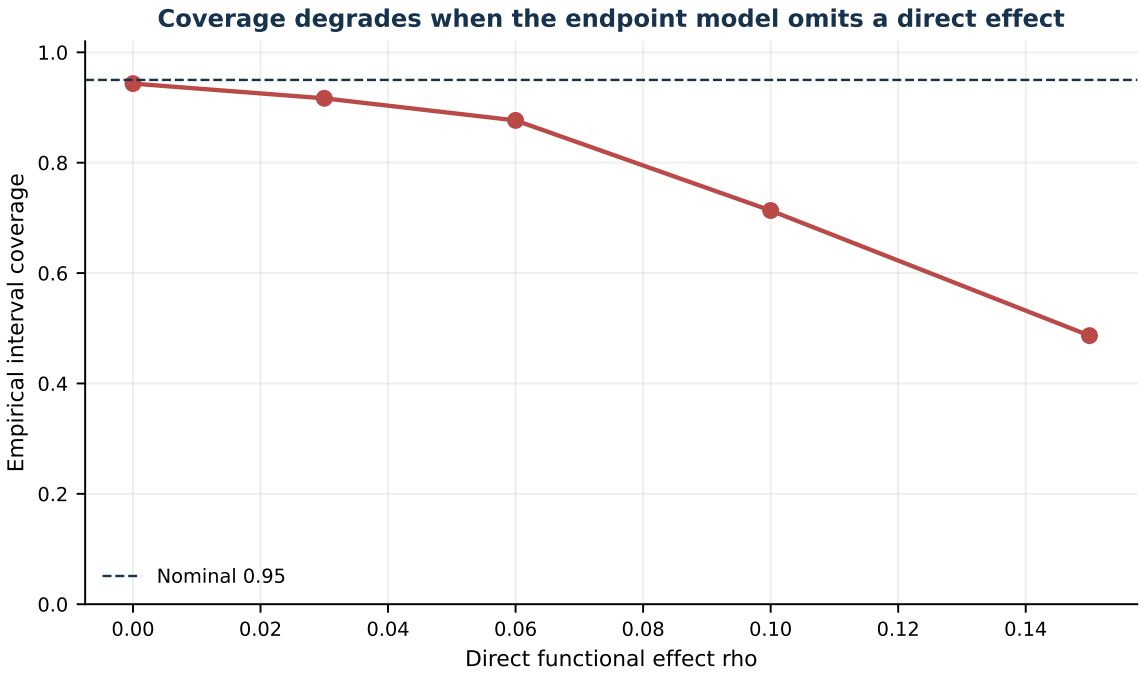


Figure 9: Approximate 95% interval coverage under increasing direct effects on the functional readout. Coverage loss is a diagnostic of model misspecification in the synthetic benchmark; it is not a biological performance estimate.

In real datasets, inherited variation is only one of several sources of heterogeneity. Cell-type mixture, batch, tissue source, medication, disease burden, and selection into a study can also shift clock values. A clock's calibration may differ across tissue or age range. The present model compresses these effects into nuisance terms to expose the identifiability issue, not to assert that all nuisance variation can be corrected by a single covariate.

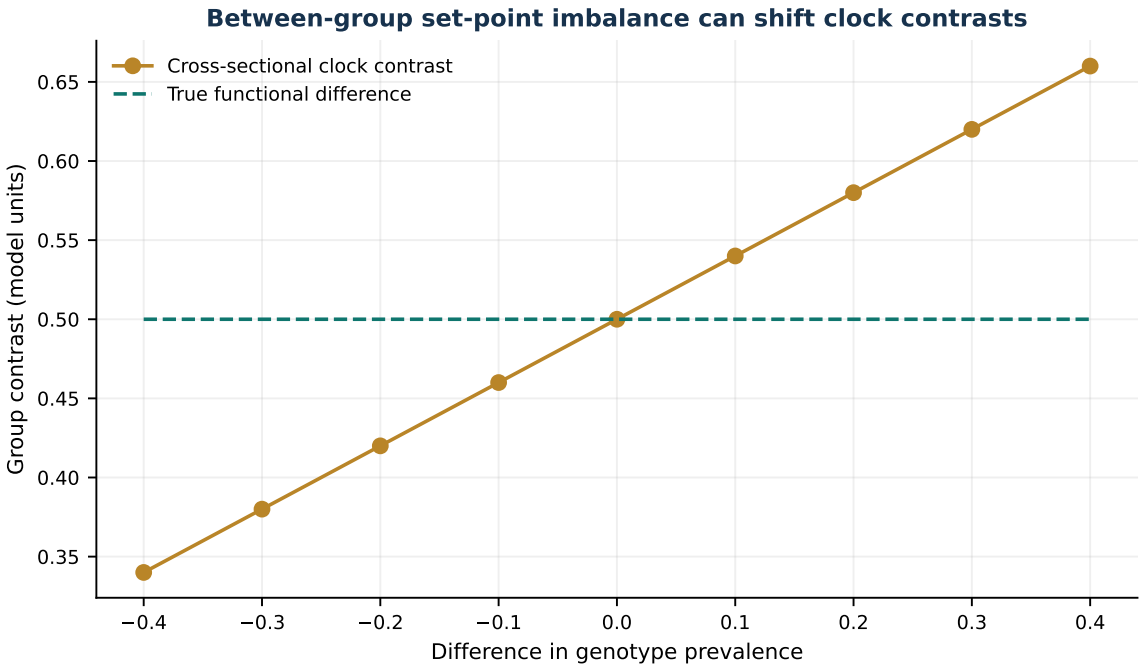


Figure 10: Sensitivity of cross-sectional clock contrasts to an additive inherited set point when group composition differs. The paired design cancels the fixed term under the model assumptions; interaction and composition changes remain outside that cancellation result.

14 Joint response surfaces and safety constraints

The response surface in fig. 11 displays clock and functional changes across combinations of the two synthetic inputs. A clock-improving region can include combinations that provide little functional benefit or carry higher identity risk. The model therefore separates efficacy-like response from safety-like burden. The risk score is illustrative and should not be interpreted as a biological safety predictor.

This separation matters for interventions intended to preserve cellular identity. The 2026 TF perturbation study explicitly searched for rejuvenating perturbations without dedifferentiation [Sengstack et al., 2026]. Our model reflects that concern as an independent coordinate rather than folding identity into the clock. The actual relation between intervention, identity, proliferation, genomic integrity, and function would need to be measured in a specific system. A favorable clock score cannot substitute for those endpoints.

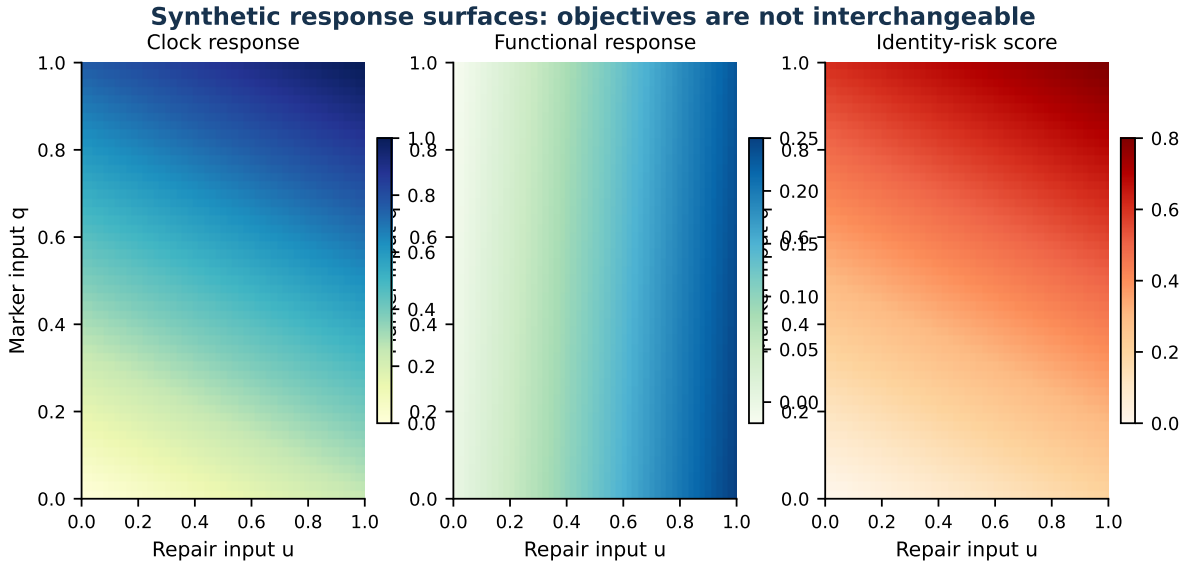


Figure 11: Synthetic response surfaces for clock improvement, functional improvement, and identity-risk burden. The surfaces are generated by the stated linear model; their purpose is to show that the objectives can disagree, not to predict any intervention's biological effect.

Table 6: Preregisterable falsification criteria for a future system-specific study.

Claim under test	Observable test	Result that weakens or falsifies the claim
Clock change reflects repair	Compare clock change with an independent functional endpoint in randomized arms.	Clock improves but the functional endpoint does not improve within prespecified uncertainty.
Marker path is separable	Vary repair and marker inputs independently and inspect design rank/conditioning.	Inputs remain collinear or the smallest singular value is effectively zero.
Functional endpoint is independent	Test marker-only controls and alternative functional readouts.	The marker-targeting arm directly shifts the endpoint or the effect reverses across readouts.
Inherited set point is controlled	Compare paired contrasts and genotype-stratified responses.	A strong genotype-by-treatment interaction or composition shift remains unexplained.
Identity is preserved	Measure cell identity and adverse outcomes separately from the clock.	Clock reduction accompanies material identity loss or adverse effects.

15 Practical identifiability: conditioning beyond rank

Structural identifiability asks whether distinct parameter values can generate exactly the same distribution of observations under an idealized design. Practical identifiability asks a harder finite-sample question: even when

the parameters are unique in principle, are the available measurements precise enough to distinguish them? These questions should not be collapsed into a single pass/fail label. A design matrix can have full column rank and still be nearly singular, so that small errors in a clock, functional assay, or intervention dose lead to large changes in the fitted coefficients.

For a locally linear model, write the stacked observations as

$$\mathbf{y} = X\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma),$$

where $\boldsymbol{\theta} = (\kappa, \alpha, \rho)^\top$, and the rows of X encode which intervention and measurement channel produced each observation. Under the stated Gaussian model, the Fisher information is

$$\mathcal{I}(\boldsymbol{\theta}) = X^\top \Sigma^{-1} X.$$

If $\mathcal{I}$ is nonsingular, the generalized least-squares covariance is $\mathcal{I}^{-1}$. Under regularity conditions, the Cramér–Rao inequality gives a lower bound on the covariance of unbiased estimators. This bound is model-conditional: it says nothing about bias introduced by an omitted causal path, a miscalibrated assay, or a functional endpoint that is not a valid proxy for the intended biological outcome.

The singular values of a suitably scaled design matrix provide a compact diagnostic. Let D be a diagonal matrix that rescales columns to comparable, prespecified units, and consider $\widetilde{X} = XD^{-1}$. If $\sigma_{\min}(\widetilde{X})$ is close to zero, there is a direction in parameter space along which predictions change very little. The condition number $\mathrm{cond}_2(\widetilde{X}) = \sigma_{\max}(\widetilde{X})/\sigma_{\min}(\widetilde{X})$ then becomes large. It is important to scale before comparing condition numbers: a change of units can alter a raw condition number without changing the underlying scientific design. We therefore report rank, scaled singular values, and the column scaling convention together.

This distinction explains why simply increasing the number of participants is not a universal remedy. Repeating the same intervention contrast can reduce the variance of the measured mean approximately as $1/n$, but it does not create a new independent direction in the design matrix. If the structural rank is deficient, arbitrarily many repetitions of the same rows leave a null space. If the design is full-rank but ill-conditioned, additional observations may help, although a new arm or independent readout can be more efficient than repeating an uninformative arm. The relevant question is not only how many observations are collected, but which causal directions those observations can separate.

A useful design report should therefore include at least four quantities: the numerical rank under a stated tolerance; the smallest singular value after prespecified scaling; a condition number or equivalent conditioning diagnostic; and the uncertainty interval for the target estimand under the fitted error model. The tolerance must be recorded because floating-point rank is not a biological property. A rank decision made with a convenient tolerance should never be presented as evidence that the underlying causal mechanism is identified.

Table 7: Interpretation of common identifiability regimes. These are mathematical design diagnostics, not empirical classifications of any biological system.

Regime	Diagnostic	What follows	Appropriate response
Structural failure	Rank deficient	At least one parameter direction is observationally equivalent under the specified model.	Add an independent perturbation or readout; do not rely on sample size alone.
Practical failure	Full rank, very small scaled singular value	Unique parameters may have unstable estimates and wide intervals.	Improve orthogonality, calibrate measurement error, or add observations in the weak direction.
Conditional adequacy	Full rank and acceptable conditioning	The declared linear parameters may be estimable under the assumed noise model.	Test model fit, residual structure, and alternative causal graphs.
Model failure	Residuals or controls contradict assumptions	Algebraic identification does not identify the intended biological mechanism.	Revise the causal model, report partial identification, or withhold the mechanistic claim.

The table is deliberately conservative. A well-conditioned design can still answer the wrong question, while a poorly conditioned design may remain useful for a coarser estimand. Before collecting data, investigators should specify which combination of parameters matters for the decision. If the target is a contrast such as

$\kappa - \rho$, it may be estimable even when every individual coefficient is not. Conversely, estimating a clock response precisely does not make a functional response identifiable. The estimand, not the convenience of the software output, should determine the design diagnostic.

16 Partial identification when direct effects cannot be calibrated

In practice, the direct effect of a marker-targeting intervention may not be known precisely. Treating an uncertain direct effect as exactly zero produces a point estimate that can look decisive while depending entirely on an untested assumption. Partial identification offers a transparent alternative: state a defensible bound on the nuisance effect and derive the range of target values compatible with the observation.

For the functional channel, suppose the observed contrast is

$$d_F = \kappa u - \rho q,$$

where $u > 0$ is the repair-labelled input, $q \geq 0$ is a marker-sensitive input, and ρ is the direct effect of the latter on the functional readout. If prior calibration or a conservative sensitivity analysis supports only $\rho \in [-R, R]$, then

$$\kappa = \frac{d_F + \rho q}{u} \quad \Longrightarrow \quad \kappa \in \left[\frac{d_F - Rq}{u}, \frac{d_F + Rq}{u} \right].$$

The interval width is $2Rq/u$. It widens linearly with the allowed direct effect and with marker-targeting dose, and it becomes unstable as u approaches zero. This is not a confidence interval: it is an identification region conditional on the stated bound. Sampling uncertainty must be layered on top of it rather than confused with it.

The same logic applies to the clock channel. If

$$d_C = \kappa u + \alpha q$$

and the direct clock effect is constrained only by $\alpha \in [-A, A]$, then

$$\kappa \in \left[\frac{d_C - Aq}{u}, \frac{d_C + Aq}{u} \right].$$

When independent functional and clock channels are available, their admissible regions can be intersected only if both equations share the same target parameter and the stated assumptions are jointly defensible. An empty intersection is informative: it indicates incompatibility among the observations, nuisance bounds, and model assumptions. It must not be repaired by silently widening bounds after seeing the result.

Proposition 3 — bounded direct-effect sensitivity

Let $u > 0$, $q \geq 0$, and $d_F = \kappa u - \rho q$. If $\rho \in [-R, R]$ for a fixed $R \geq 0$, the set of values of κ compatible with the exact contrast is the interval $[(d_F - Rq)/u, (d_F + Rq)/u]$. Its width is $2Rq/u$. If no finite bound on ρ is available and $q > 0$, the contrast alone does not provide a finite identification interval for κ.

The result follows directly by solving the contrast for κ and taking the image of the closed interval $[-R, R]$ under the affine map $\rho \mapsto (d_F + \rho q)/u$. It is an algebraic statement, not a claim that any particular biological direct effect lies within a chosen bound.

The practical value of the bound is that it makes assumptions visible. A sensitivity analysis can vary R, plot the resulting identification region, and report the threshold at which the sign of κ becomes indeterminate. If a proposed benefit is positive only when $|\rho|$ is assumed to be tiny, the conclusion is fragile. If the sign remains positive across a scientifically defensible range, the evidence is more robust to that specific uncertainty—but remains conditional on the linear model, the endpoint definitions, and the absence of other omitted paths.

Bounds should be chosen before examining the target outcome where possible. They may be informed by independent calibration, negative controls, assay validation, prior studies, or explicit expert elicitation. If no credible bound can be defended, the correct result may be a wide or unbounded identification region. That is a substantive finding about what the current design can establish, not a failure of analysis.

17 Repeated measures, batch effects, and latent composition

Paired measurements are attractive because they remove stable subject-specific baselines. Yet a within-subject difference is not automatically causal. Time-varying confounding, regression to the mean, batch shifts, differential cell survival, and changes in cell-type composition can still alter the observed contrast. The distinction matters especially when the outcome is a molecular score constructed from features that may themselves respond directly to the intervention.

A simple hierarchical extension is

$$Y_{ijt} = \mu + \beta_u u_{it} + \beta_q q_{it} + b_i + c_j + \delta_t + \epsilon_{ijt},$$

where b_i is a subject-level random intercept, c_j is a batch effect, and δ_t represents a time-specific influence shared across samples. This expression is a bookkeeping model, not a universal biological law. If each intervention is confounded with a unique batch, the corresponding effects cannot be separated by fitting a more elaborate mixed model alone. The design must contain overlap: multiple arms represented within batches, randomized processing order where feasible, and technical controls that reveal assay drift.

Cell composition creates a second layer of ambiguity. Let the measured bulk clock be a weighted aggregate of cell-state-specific values, $C_{\mathrm{bulk}} = \sum_k \pi_k C_k$, where π_k is the proportion of state k. A change in C_{bulk} can arise from within-state molecular change, a shift in the π_k, or both. These possibilities have different interpretations. A treatment that selectively removes a high-clock subpopulation may lower the bulk score without resetting the clock in surviving cells. Conversely, a beneficial within-cell change can be obscured by a shift in composition. Bulk measurements alone do not generally distinguish these paths.

The decomposition is not solved by adding a composition covariate after the fact if the intervention itself changes composition and composition lies on a causal pathway of interest. Adjusting for a post-intervention variable can change the estimand or introduce collider bias under some causal structures. Investigators should specify whether they seek a total intervention effect, a within-cell effect, a composition-standardized contrast, or a decomposition into both components. These are different questions and may require different measurements or assumptions.

A robust protocol should therefore combine paired baselines with randomized intervention assignment, balanced batches, explicit composition measurements, and independent functional outcomes. When single-cell assays are appropriate and feasible, cell-resolved data can help distinguish within-state changes from shifts in population proportions; however, single-cell data introduce their own sampling, dropout, dissociation, and annotation uncertainties. More granular data are not automatically more informative if the measurement process is poorly calibrated or the relevant cell states are not identifiable.

Negative controls are especially valuable when they target a plausible measurement path without being expected to repair the function of interest. A marker-sensitive control tests whether the clock can move in the absence of the proposed repair pathway. A sham or vehicle control estimates temporal drift and handling effects. An orthogonal functional assay tests whether the response generalizes beyond the original measurement channel. The controls must be chosen for the actual system; the names of the arms do not guarantee that their causal effects match the intended assumptions.

18 Sequential design under model uncertainty

The finite-candidate planner in this release is intended to make experimental choices auditable. Its objective can be written abstractly as

$$J(A) = \log\det\left(\mathcal{I}_0 + \sum_{a\in A} X_a^\top \Sigma_a^{-1} X_a\right) - \lambda\,\mathrm{Cost}(A) - \mu\,\mathrm{Risk}(A),$$

where A is a set of candidate arms, $\mathcal{I}_0$ is prior information, and the final two terms encode resource use and prespecified safety constraints. The log-determinant rewards designs that improve information across multiple parameter directions rather than concentrating all information on one coefficient. It is one defensible criterion, not a universally optimal objective.

In a Bayesian formulation, an alternative is expected information gain,

$$\mathrm{EIG}(a) = \mathbb{E}_{y_a \mid D,a}[D_{\mathrm{KL}}\{p(\theta \mid D, y_a, a) \,\|\, p(\theta \mid D)\}] ,$$

where D is the current evidence and a is a proposed next arm. This criterion asks how much the next observation is expected to change the parameter distribution. It can be useful when uncertainty is non-Gaussian or model-averaged, but it is sensitive to the prior, likelihood, candidate set, and computational approximation. A high expected information gain under a misspecified prior can direct attention toward the wrong scientific distinction.

For a safety-constrained problem, the optimization should be lexicographic. First reject arms that violate hard identity, viability, or ethical constraints. Then require that the proposed design identifies the stated estimand under the declared model. Only then optimize information per unit resource. A penalty term is not always an adequate substitute for a hard constraint: if a risk score is poorly calibrated, a large information reward can overwhelm the penalty. The implementation in this paper is a synthetic design exercise and does not contain a validated biological risk model.

Sequential choice also creates selection effects. If the next arm is chosen using interim outcomes, the selection rule is part of the data-generating process and should be recorded. Repeatedly inspecting a noisy endpoint and stopping when the clock moves in the desired direction can inflate false-positive evidence. A valid adaptive protocol should prespecify the candidate library, update rule, stopping criterion, and final inferential procedure. Where conventional frequentist guarantees are desired, the analysis must account for adaptive sampling; where Bayesian inference is used, the prior and likelihood must be reported and calibration checked.

Greedy design is computationally convenient, but it need not achieve the globally best set when candidates interact or when constraints are non-submodular. The release therefore treats the selected arm set as a proposal with a traceable score, not as a proof of global optimality. A future implementation could compare greedy selection with exhaustive search on small candidate sets, mixed-integer optimization, or posterior-simulation methods. Such comparisons would test the planner itself rather than assuming that a plausible-looking output is optimal.

19 Sample-size logic and preregistered estimands

Sample size should be tied to the estimand and the weakest design direction, not to the desire for a narrow interval around a convenient clock score. In a correctly specified linear model with independent homoscedastic errors, information grows approximately linearly with replicated observations, so standard errors often decline on the order of $n^{-1/2}$. This familiar scaling does not resolve structural non-identifiability, and it can be misleading when errors are correlated, measurement variance depends on the state, or the design is selected adaptively.

A useful planning exercise begins with a pilot-informed range for measurement variance and direct-effect sensitivity. For each candidate sample size, simulate the full planned design, including missingness, batch allocation, baseline correlation, and any sequential decision rule. Record bias, root-mean-square error, interval coverage, false attribution of clock-only movement to repair, and the probability that a safety or identity constraint is triggered. The target should be a decision-relevant operating characteristic—for example, the probability that the identification region excludes a prespecified null range—rather than a single nominal power number disconnected from model uncertainty.

The synthetic simulations in this paper are not a biological power calculation. Their parameters are chosen to expose a mathematical failure mode and to verify the software path. They do not estimate the variance of a real methylation assay, the prevalence of cell-state transitions, or the distribution of intervention effects in humans or model organisms. Any real sample-size recommendation would require pilot data, a specific assay, a declared population, and a defensible model for missingness and heterogeneity.

Preregistration should name the primary clock, independent functional endpoint, identity and safety outcomes, contrast of interest, nuisance-effect bounds, exclusions, missing-data handling, and stopping rule. It should also state which outcomes are confirmatory and which are exploratory. If multiple clocks or functional endpoints are measured, multiplicity and endpoint selection must be addressed. Selecting the clock that improves most after observing the data is not equivalent to testing a prespecified biological hypothesis.

The report should preserve negative results and discordant outcomes. If the clock improves but function does not, the result is not a failed rejuvenation experiment that can be rescued by re-labeling the clock; it is evidence against the specific claim that the observed clock movement establishes functional repair. If function

improves while the clock does not, the clock may be insensitive to that mechanism. If both improve but identity or safety worsens, the intervention cannot be described as an unqualified success. These branches should be written into the analysis plan before data collection.

20 Stress tests and the boundary of interpretation

A mathematical model is most useful when it identifies where it can fail. The linear contrast model assumes that the effects of u and q are additive over the tested range. Saturation, thresholds, dose interactions, delayed effects, and state-dependent sensitivity can violate that assumption. A nonlinear model may be appropriate, but adding nonlinear terms can also consume the independent variation required to identify them. The correct response is not to fit the most flexible model automatically; it is to ask whether the design contains enough information to distinguish the extra mechanism.

A first stress test is an interaction model, $d_C = \kappa u + \alpha q + \gamma uq + \epsilon_C$. If all observations use the same ratio of u to q, the interaction may be nearly collinear with the main effects. A second stress test allows the direct marker effect to vary across subjects or batches, $\alpha_i = \alpha + \eta_i$. A third introduces delayed response, replacing u_t by a distributed-lag term $\sum_{\ell=0}^{L} h_\ell u_{t-\ell}$. Each extension changes the design requirements. The parameters cannot be declared identifiable merely because an optimizer returns finite estimates.

Measurement error in the intervention itself is another important threat. If the nominal dose differs from the delivered exposure, regression coefficients can be attenuated or distorted. When dose measurement error is correlated with batch or subject state, the bias may be more complicated than simple attenuation. Calibration samples, randomized assignment, and independent exposure measures may be needed. A model that treats the intervention input as known should explicitly state that assumption.

The causal graph can also be wrong. The proposed repair-labelled input may affect the clock through a direct molecular path; the marker-targeting input may affect function; both inputs may change cell composition; and an unmeasured stress response may affect all readouts. No rank condition eliminates these alternatives. They require negative controls, independent readouts, intervention-specific knowledge, or a sensitivity analysis that bounds the unmeasured paths. If no credible bound or control exists, the mechanism should remain unresolved.

These stress tests motivate a tiered interpretation. A result can establish a mathematical identity in the synthetic model; demonstrate that the software recovers a known synthetic parameter under stated assumptions; show robustness to a declared range of nuisance effects; or support a causal biological claim in a properly designed experiment. These are four different evidentiary levels. The current release reaches only the first three within a synthetic setting, and it does not claim the fourth.

The most decisive falsification is not a small change in one metric. It is a pattern that contradicts the central measurement claim: a clock-only perturbation produces the expected clock response but no corresponding independent functional response; the inferred repair coefficient changes sign under modest, independently plausible direct-effect bounds; or the proposed model fails to predict held-out functional outcomes. Each outcome would require the mechanism claim to be narrowed, revised, or withheld. A preregistered negative result can be more informative than an apparently large clock reset whose causal interpretation is not identified.

21 Counterexamples and failure modes

The model generates several counterexamples to a simplistic clock-only interpretation. First, a marker-only intervention can change C while $\kappa = 0$. Second, a genuine repair intervention can be obscured if the direct marker effect has the opposite sign. Third, a functional readout can itself be directly affected by a marker-targeting intervention, violating the assumption that it is an independent endpoint. Fourth, a genotype-by-treatment interaction can survive within-subject differencing. Fifth, a well-conditioned design can still estimate the wrong parameters if the observation equations omit a relevant path.

These counterexamples are not edge cases added after the fact; they define the assumptions under which the method is interpretable. Table 6 converts the central claims into preregisterable tests, while Table 8 summarizes failure signatures and required responses. A useful validation workflow should actively seek such failures. For each proposed intervention, the experimenter should specify at least one marker-only negative control, one functional endpoint independent of the clock-training features, an identity/safety panel, and a perturbation dose or timing contrast that can reveal direct measurement effects.

Table 8: Failure modes, observable signatures, and required responses.

Failure mode	Observable signature	Required response
Direct marker shift	Clock improves but independent function does not.	Do not attribute the full clock change to repair.
Functional endpoint direct effect	Functional contrast changes in marker-only control.	Estimate or bound ρ; add another independent endpoint.
Genotype-by-treatment interaction	Paired effects vary by genotype stratum.	Report interaction and stratum-specific uncertainty.
Composition shift	Clock changes with cell-type proportions.	Measure composition or use cell-resolved analysis.
Ill-conditioned design	Small singular value / large condition number.	Add orthogonal perturbation arms or readouts.
Identity loss	Clock improves while identity or safety measures worsen.	Treat as a separate adverse outcome; do not optimize the clock alone.

22 A falsifiable experimental protocol

A future biological study could test the model only after selecting a specific cell system, intervention, and independently justified functional endpoint. We do not prescribe molecular targets or treatment parameters. The protocol below is a design template: preregister the causal question, identify the proposed repair and marker-sensitive paths, define the estimand and safety limits, and choose a measurement schedule that permits the two paths to be separated.

The design should include untreated or vehicle controls, a marker-targeting control expected to move the clock without the proposed repair mechanism, and at least one perturbation expected to change function through an independently supported pathway. Randomize intervention components where scientifically and ethically appropriate. Measure the clock, a functional outcome, cell identity, and relevant adverse outcomes at baseline and follow-up. If the system is heterogeneous, use stratification or cell-resolved measurements and record composition changes. The analysis should report both the total clock response and the functional response rather than converting one into the other.

The primary falsification criterion is straightforward: if a clock-lowering perturbation fails to improve the preregistered functional endpoint, or if the effect disappears after accounting for direct marker changes, the study does not support the claim that the clock reduction indicates functional repair. A second criterion is failure of the model's predicted dose-response or interaction structure. A third is a meaningful increase in identity loss or adverse outcomes, even if the clock improves. Null or adverse results are informative and must be retained.

23 An explicit decision rule

The output of the planner is not a single rejuvenation score. It is a recommended set of admissible arms, their predicted information contribution, the rank and conditioning of the resulting design, and a list of unresolved assumptions. For a proposed biological application, the decision rule should be lexicographic: first satisfy prespecified safety and identity constraints; next require the design to identify the target estimand under the declared model; then maximize information per unit cost. A design that violates a hard safety constraint is not rescued by a high information score.

The synthetic implementation uses a finite candidate library and greedy addition. At each step it evaluates the log-determinant gain from each admissible candidate, subtracts a cost penalty, rejects candidates above the risk ceiling, and chooses the best remaining candidate. The algorithm is auditable and deterministic for a fixed seed. It is not a substitute for domain expertise, and its safety score is only as valid as the assumptions supplied to it.

The design objective can be extended to incorporate prior uncertainty over parameters, expected model discrimination, missingness, batch effects, and multiple tissue endpoints. Such extensions are worthwhile only if the additional structure is estimable and validated. A complex planner can be confidently wrong when its prior or risk model is misspecified. For this reason, we archive candidate scores, constraints, intermediate information matrices, and the selected design rather than reporting only the final choice.

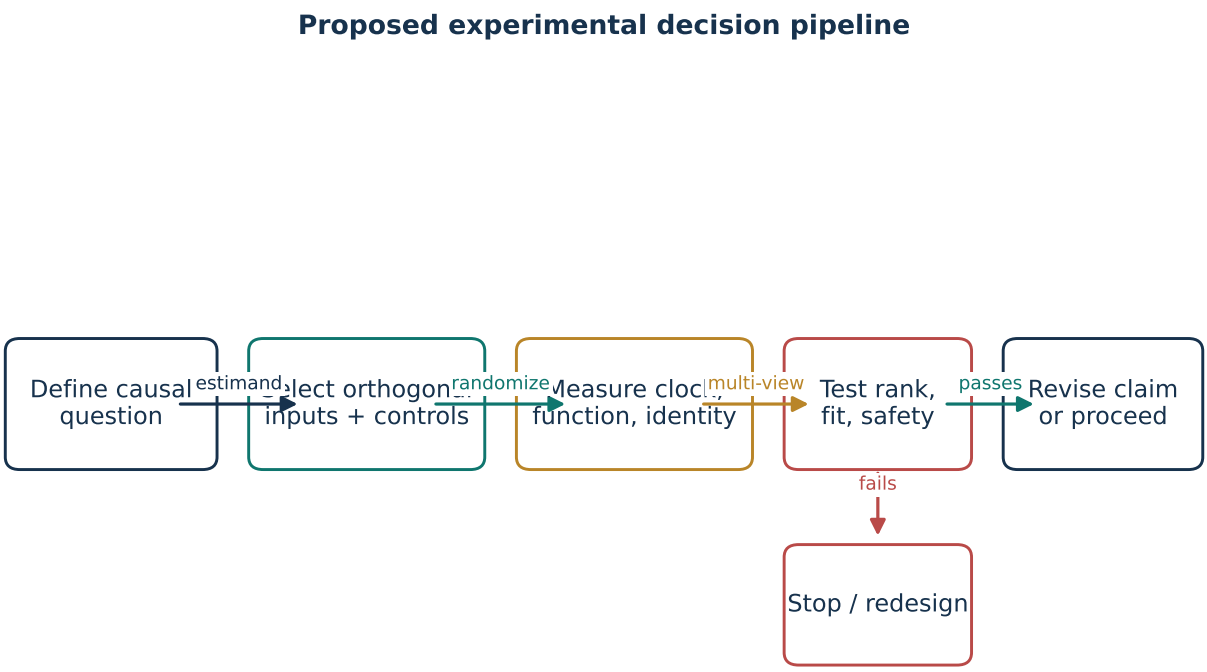


Figure 12: Proposed validation workflow. The stages are a research-design template, not a completed experiment. A study should stop or revise the claim if functional, identity, or safety endpoints contradict the clock-based interpretation.

24 Ethical and translational boundaries

Aging interventions and cellular reprogramming have potential relevance to healthy lifespan, regenerative medicine, and treatment of age-associated disease, but the gap between a synthetic design model and a biological intervention is substantial. A false claim of rejuvenation could misdirect resources, obscure adverse effects, or encourage premature translation. This paper therefore makes no clinical recommendation, no claim of lifespan extension, and no assertion that a clock reset is inherently beneficial or harmful.

A valid application would require domain-specific safety review, suitable cell or organism models, preregistered outcomes, appropriate controls, and ethical/regulatory oversight. For cellular reprogramming, preservation of identity and avoidance of tumorigenic or dysplastic states are separate concerns from molecular age markers. A lower score should never be treated as a safety guarantee. The proposed active-design planner should prioritize the safest admissible experiments rather than maximize a surrogate endpoint without constraints.

The AI component also has limits. The planner does not infer causal biology from unverified data, and it does not independently assign a molecular target to an intervention. Its role is to expose which experiment would be informative under an explicit model. Human investigators must decide whether the model and candidate interventions are scientifically plausible, ethically acceptable, and adequately controlled.

25 Discussion

The main result is a measurement-identifiability statement, not a new theory of aging. If a clock can respond to direct marker perturbation, a clock-only outcome cannot generally identify functional repair. This is consistent with the broader distinction between prediction and causal intervention. The recent literature gives concrete reasons to model direct methylation effects and genetic regulation explicitly [Liesenfelder et al., 2025, Lim, 2026]. It also illustrates that intervention discovery can be combined with identity constraints rather than using an age score as the sole objective [Sengstack et al., 2026].

The proposed response is methodological. Treat the clock as one measurement channel among several; separate within-subject change from baseline between-person differences; randomize intervention components where possible; include an independent functional endpoint and an identity/safety panel; and report rank, conditioning, uncertainty, and counterexamples. If an experiment cannot distinguish the proposed paths, the correct conclusion is that the mechanism remains unidentified, not that more subjects will necessarily resolve it.

There are limits to the linear model. Biological systems can exhibit nonlinear response, delayed adaptation, feedback, threshold effects, cell-state mixtures, and intervention-specific changes in the measurement process. A scalar latent functional coordinate is itself a simplification. In future work, the model could be extended to a partially observed dynamical system with time-varying coefficients and multiple endpoints. But additional complexity should be earned by data and identifiability analysis. A high-dimensional model can conceal rather than solve the problem if its parameters cannot be distinguished by the experiment.

26 Limitations

First, the benchmark is synthetic and uses known linear equations. It demonstrates an identifiability failure that follows from those equations, not the frequency or magnitude of the failure in real rejuvenation experiments. Second, the functional endpoint is assumed to have a known interpretation and a simple direct-effect term. Real functional measures may be noisy, tissue-specific, and causally affected by multiple intervention paths. Third, the genotype model is additive and time-invariant; interactions, ancestry structure, and cell composition require richer models and better data.

Fourth, the active-design method optimizes a finite candidate set using a local Fisher-information approximation and a stylized risk penalty. It does not guarantee global optimality for nonlinear models, and it cannot make unsafe candidate interventions safe. Fifth, we did not analyze a public methylation dataset or reproduce any published experiment. The literature review was focused, not exhaustive, and the topic-selection scores are internal triage judgments rather than validated metrics of novelty or impact. Finally, no claim is made that the proposed design rule has been implemented or prospectively tested in a biological laboratory.

27 Conclusion

A change in an epigenetic clock is not, by itself, an identified change in functional aging. In a minimal state-space model, a scalar clock contrast can be generated by multiple combinations of latent repair and direct marker displacement. This is a structural ambiguity, not merely a small-sample problem. Paired contrasts can remove an additive inherited set point under stated assumptions, while independent perturbation variation and functional measurements can improve identifiability.

The practical output is a falsifiable design rule: plan experiments to separate the repair path from the measurement path, report the rank and conditioning of the design, and keep identity and safety as independent outcomes. The synthetic benchmark demonstrates how a clock-only attribution rule can fail and provides code to reproduce the result. Whether this framework improves real biological experiments remains an open empirical question.

Acknowledgments and disclosure

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A Proof of Proposition 1

For fixed nonzero u, q, the expected scalar clock contrast is $y = \kappa u + \alpha q$. For any real t, define $\kappa' = \kappa + tq$ and $\alpha' = \alpha - tu$. Then

$$\kappa' u + \alpha' q = (\kappa + tq)u + (\alpha - tu)q = \kappa u + \alpha q + t(qu - uq) = \kappa u + \alpha q.$$

Thus a continuum of parameter pairs yields the same expected observation. The pair is not uniquely identified by y. The argument concerns the expectation; adding zero-mean noise changes uncertainty but does not remove the structural equivalence.

B Proof sketch for Proposition 2

Each clock observation contributes a row $(u, q, 0)$ and each functional observation contributes a row $(u, 0, -q)$ to the Jacobian with respect to (κ, α, ρ). Local identification of a linear model requires the Jacobian to have full column rank. If all arms use a single fixed pair (u, q), the stacked matrix contains at most two distinct row

directions and has rank at most two. Independent variation in u and q across arms, with both observation channels present, can generate three linearly independent columns. The exact rank depends on the selected design; it must be computed, not assumed.

C Active-design algorithm

1. Define a finite set of admissible input pairs (u, q), measurement channels, costs, and a prespecified identity-risk score.
2. For each candidate, construct the observation Jacobian and the corresponding Fisher-information contribution under the stated noise covariance.
3. At each sequential step, compute the increase in regularized log-determinant for every candidate that remains within the cost and risk budgets.
4. Select the candidate with the largest admissible information gain; record its score, constraints, and rejected alternatives.
5. After the planned allocation is selected, report the unregularized design rank, singular values, condition number, and sensitivity to the risk model.

The released implementation uses a deterministic finite candidate library. It is a transparent greedy heuristic, not a claim of global optimality.

D Synthetic parameter register

The benchmark uses a fixed seed, dimensionless coefficients, Gaussian measurement noise, and repeated simulated subjects. The complete parameter values and dependency versions are written by the script to `results/summary.json` and `results/environment_report.json`. All figures are regenerated from the same model code. No empirical dataset is downloaded or required.

E Reproducibility and interpretation

Run `python verify_model.py` from the release directory. The script generates the synthetic source data, model checks, figure assets, figure inventory, candidate-topic scores, literature audit, and generated LaTeX macros. Run `python deploy.py -self-test` to test fail-closed behavior on an intentionally incomplete fixture. Run `python deploy.py -build` to regenerate the figures, compile the current manuscript, and run automated preflight. The publication pipeline records the PDF hash and renders all pages. Manual visual inspection is recorded separately; automated checks do not substitute for scientific or visual review.